



Important Formulas & Points to Remember: Constructions and Tilings

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Symmetry Condition in Eye Construction

For the eye shape to be **perfectly symmetrical**, the following distances must all be equal:

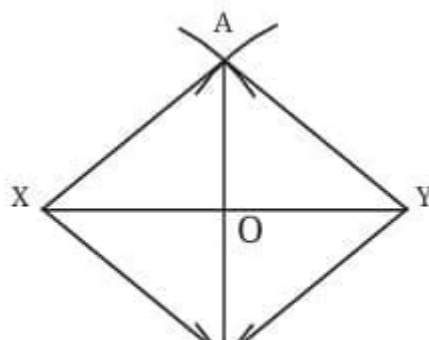
$$AX = AY = BX = BY$$

Why?

A and B are the centres of the upper and lower arcs.

X and Y lie at the ends of the supporting line.

Equal distances ensure **equal radii** for the upper and lower arcs → perfect symmetry.





Resulting Geometric Facts:

Line **AB** meets **XY** at its midpoint **O**.

Line **AB** $\perp$ **XY** (AB is perpendicular to XY).

Theorem

Statement:

Any point with equal distance from X and Y must lie on the **perpendicular bisector of XY**.

Why?

A line segment has only **one** perpendicular bisector.

Points satisfying **PX = PY** for a segment XY always align on this single line.

Construction: Perpendicular Bisector of a Line Segment XY

Steps

1. With X and Y as centres, draw two arcs **above** XY using the same radius $\rightarrow$ meet at A.
2. Draw two arcs **below** XY with the same radius $\rightarrow$ meet at B.
3. Join A and B.

AB = Perpendicular bisector of XY

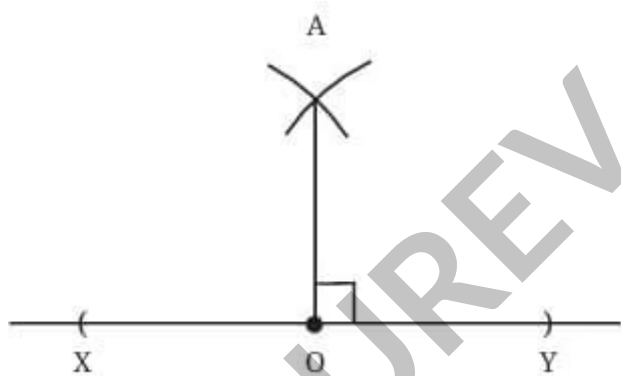
Why this works?

A and B are **equidistant** from X and Y $\rightarrow$ both lie on the perpendicular bisector line.

Construction: Right Angle (90°) at a Point O on a Line

Steps

1. Choose X and Y on the line such that **OX = OY**.
2. Construct the perpendicular bisector of XY (need only one point above the line).
3. Join that point with O $\rightarrow$ gives a **90° angle** at O.

**Key Idea:**

The perpendicular bisector of a segment always makes a **right angle** with the segment.

Formula for Dividing a Circle into Equal Angles

$$\text{Angle between lines} = \frac{360^\circ}{n}$$

Construction of a Parallel Line Using Only Compass & Ruler

Key Idea:

Parallel lines have **equal corresponding angles**.

Steps:

1. Draw a transversal l intersecting the line m at point A .
2. Choose point B (where parallel line should pass).
3. Copy the angle at A to point B .
4. Draw a line through B that forms the copied angle $\rightarrow$ this line $\parallel m$.

Result:

A new line parallel to m is constructed **without a set square**.

